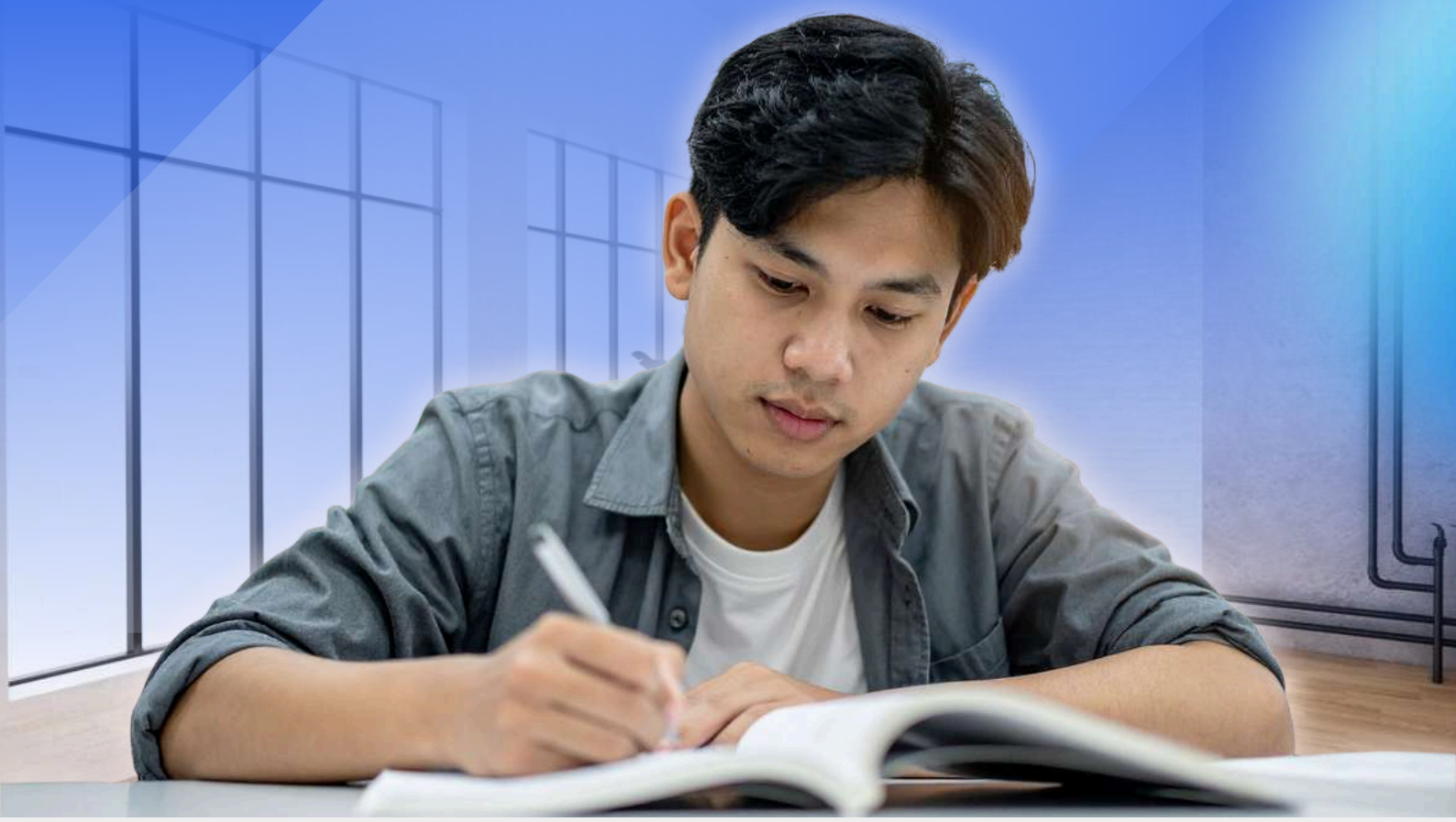
Lenterago

TRY OUT #9

SAT-Math

61009

44 QUESTIONS



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MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

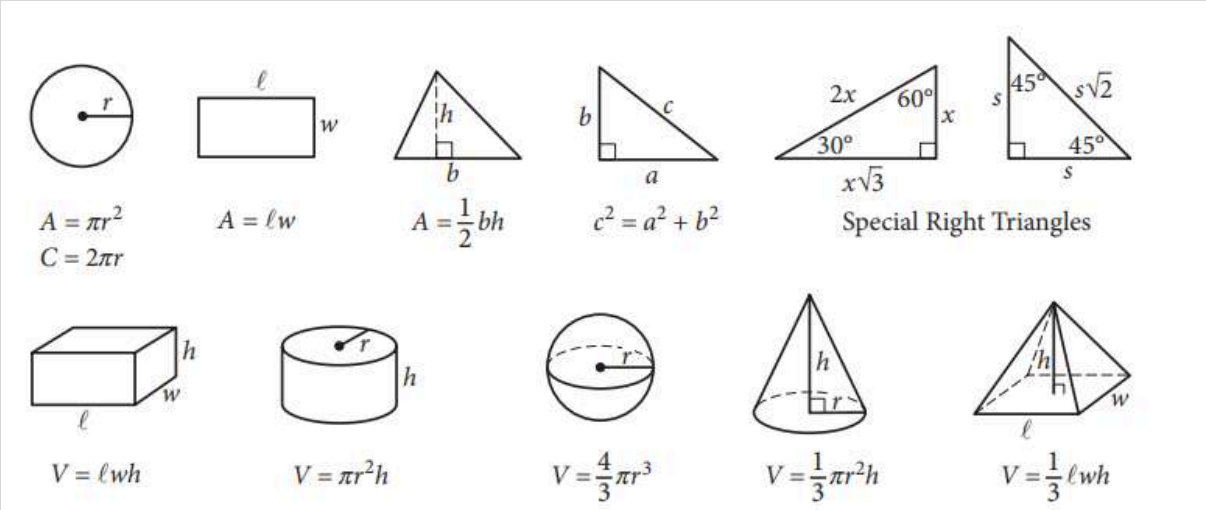
A. All variables and expressions represent real numbers.

B. Figures provided are drawn to scale.

C. All figures lie in a plane.

D. The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

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Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

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1. A principal used a total of 25 flags that were either blue or yellow for field day. The principal used 20 blue flags. How many yellow flags were used?
   A. 5
   B. 20
   C. 25
   D. 30

2. $$a(3 - x) - b = - 1 - 2x$$
   In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?
   A. $a = 2$ and $b = 1$
   B. $a = 2$ and $b = 7$
   C. $a = - 2$ and $b = 5$
   D. $a = - 2$ and $b = - 5$

3. A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.
   $$y = 120 - 25x$$
   The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?
   A. The team will have moved all the cargo in about 4.8 hours.
   B. The team has been moving about 4.8 tons of cargo per hour.
   C. The team has been moving about 25 tons of cargo per hour.
   D. The team started with 120 tons of cargo to move.

4. The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x-coordinate of the x -intercept of the graph of $y = h(x)$ in the xy-plane?

5. On a 210 -mile trip, Cameron drove at an average speed of 60 miles per hour for the first x hours. He then completed the trip, driving at an average speed of 50 miles per hour for the remaining y hours. If $X = 1$, what is the value of y ?

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6. $$4x - 6y = 10y + 2$$
$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

7. $$y > 7x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	13
5	27
8	48

B.

x	y
3	17
5	31
8	52

C.

x	y
3	21
5	27
8	52

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D.

x	y
3	21
5	35
8	56

8. A team hosting an event to raise money for new uniforms plans to sell at least 140 tickets before this event and at least 220 tickets during this event to raise a total of at least $\$5,820$ from all tickets sold. The price of a ticket during this event is $\$3$ less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?
   A. $140(x + 3) + 220x \leq 5,820$
   B. $140(x + 3) + 220x \geq 5,820$
   C. $140(x - 3) + 220x \leq 5,820$
   D. $140(x - 3) + 220x \geq 5,820$

9.

x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y. Which of the following graphs in the xy-plane could represent the relationship between x and y ?

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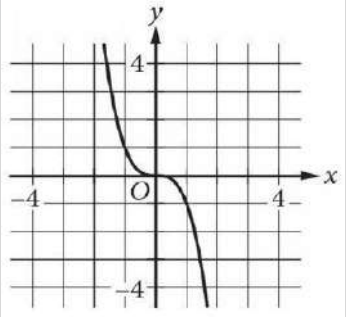
A.



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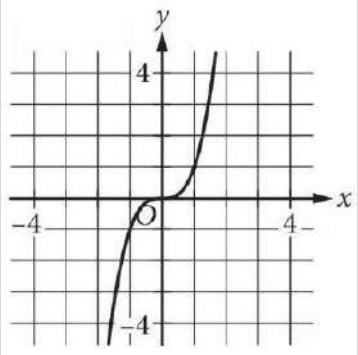

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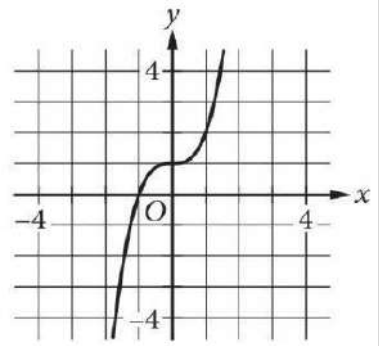

D.

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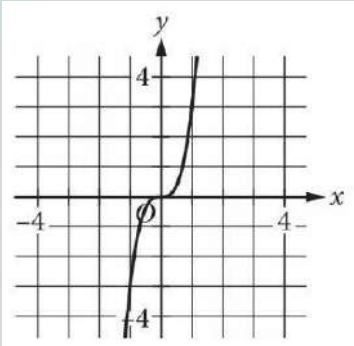

10. A rectangle has a length of x units and a width of ($x - 15$) units. If the rectangle has an area of 76 square units, what is the value of x ?
    A. 4
    B. 19
    C. 23
    D. 76

11. $$(x - 4)(x + 2)(x - 1) = 0$$

    What is the product of the solutions to the given equation?

    A. 8
    B. 3
    C. -3
    D. -8

12. In the xy-plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

13. If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?
    A. $3x^2 + 12x + 7$
    B. $3x^2 + 14x + 17$
    C. $3x^2 + 19x + 20$
    D. $3x^2 + 26x + 33$

14. $$(ax + 3)\left(5x^2 - bx + 4\right) = 20x^3 - 9x^2 - 2x + 12$$

    The equation above is true for all x, where a and b are constants. What is the value of ab?

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A. 18
B. 20
C. 24
D. 40

15. For a person m miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is k seconds. The ratio of m to k can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?
A. 10
B. 9
C. 6
D. 5

16. Rectangle A has length 15 and width w. Rectangle B has length 20 and the same length-to-width ratio as rectangle A. What is the width of rectangle B in terms of w ?
A. $\frac{4}{3}w$
B. $w + 5$
C. $\frac{3}{4}w$
D. $w - 5$

17. A school district is forming a committee to discuss plans for the construction of a new high school. Of those invited to join the committee, 15% are parents of students, 45% are teachers from the current high school, 25% are school and district administrators, and the remaining 6 individuals are students. How many more teachers were invited to join the committee than school and district administrators?

18. Which of the following is true about the values of 2^x and $2x + 2$ for $x > 0$?
A. For all $x > 0$, it is true that $2^x < 2x + 2$.
B. For all $x > 0$, it is true that $2^x > 2x + 2$.
C. There is a constant c such that if $0 < x < c$, then $2^x < 2x + 2$, but if $x > c$, then $2^x > 2x + 2$.
D. There is a constant c such that if $0 < x < c$, then $2^x > 2x + 2$, but if $x > c$, then $2^x < 2x + 2$.

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19.

	Blood type			
Rhesus factor	A	B	AB	O
+	33	9	3	37
-	7	2	1	x

Human blood can be classified into four common blood types- A, B, AB, and O. It is also characterized by the presence $(+)$ or absence $(-)$ of the rhesus factor. The table above shows the distribution of blood type and rhesus factor for a group of people. If one of these people who is rhesus negative $(-)$ is chosen at random, the probability that the person has blood type B is $\frac{1}{9}$. What is the value of x ?

20. Point F lies on a unit circle in the xy-plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH, in radians?

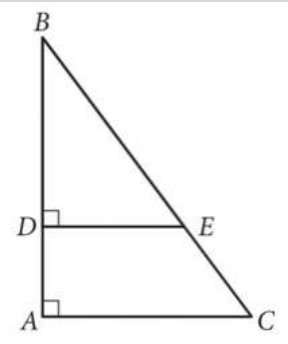
A. $\frac{27\pi}{2}$

B. $\frac{29\pi}{2}$

C. 24π

D. 25π

21.



In the figure above, $tanB = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

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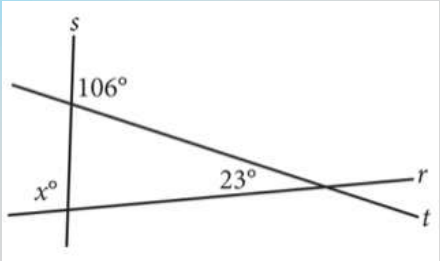
22. Intersecting line r, s, and t are shown below.

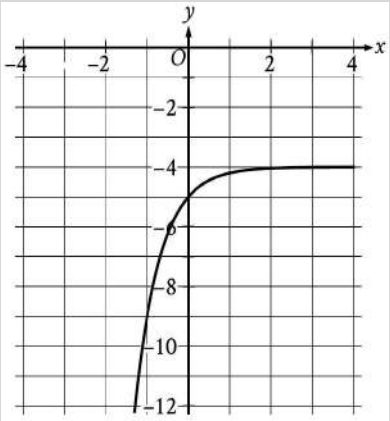


What is the value of x?

23. If $3x - 8 = 7$, what is the value of $3x + 8$?

A. -1
B. 5
C. 13
D. 23

24.



What is the y-intercept of the graph shown?

A. $(-1, -9)$
B. $(0, -5)$
C. $(0, -4)$
D. $(0, 0)$

25. $$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

A. Exactly one
B. Exactly two
C. Infinitely many
D. Zero

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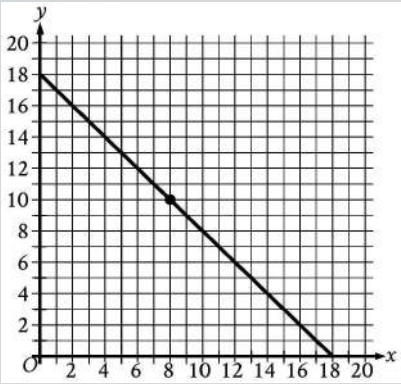
26.



The graph in the xy-plane models the possible combinations of length x, in meters (m), and width y, in meters, for a rectangle with a perimeter of 36 m. Which statement is the best interpretation of the point $(8, 10)$ in this context?

A. The length is 10 m less than the perimeter, and the width is 8 m less than the perimeter.
B. The length is 10 m, and the width is 8 m .
C. The length is 8 m, and the width is 10 m .
D. The length is 8 m less than the perimeter, and the width is 10 m less than the perimeter.

27.

$$-x + y = -3.5$$

$$x + 3y = 9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

28. Which of the following expressions is (are) a factor of $3x^2 + 20x - 63$?

I. $x - 9$
II. $3x - 7$

A. I only
B. II only
C. I and II
D. Neither I nor II

29. A right circular cylinder has a volume of 45π. If the height of the cylinder is 5, what is the radius of the cylinder?

A. 3
B. 4.5
C. 9
D. 40

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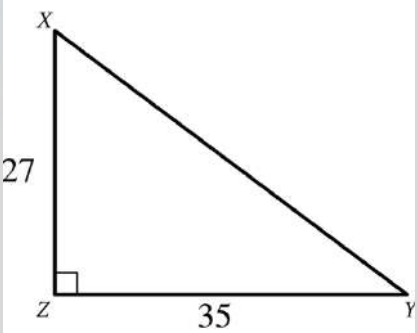
30.



Note: Figure not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $sinX$?

A. $tanX$
B. $tanY$
C. $cosX$
D. $cosY$

31. $$\frac{12x+28}{4} - \frac{s}{13} = r(x - 8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

32. According to a model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to four times the body weight of the bee, in grams. According to the model, what would be the head width, in millimeters, of a worker bumblebee that has a body weight of 0.5 grams?

33. $$y = x + 1$$

$$y = x^2 + x$$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

A. -1
B. 0
C. 2
D. 3

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34.

x	y
-12	-45
6	45

The table shows two values of x and their corresponding values of y. The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{4}, a\right)$. What is the value of a ?

35. A cargo helicopter delivers only 100-pound packages and 120-pound packages. For each delivery trip, the helicopter must carry at least 10 packages, and the total weight of the packages can be at most 1,100 pounds. What is the maximum number of 120-pound packages that the helicopter can carry per trip?

A. 2
B. 4
C. 5
D. 6

36. $$f(x) = 4000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \leq 7$. Which of the following is the best interpretation of 4000 in this context?

A. The estimated initial beetle population for this species and area in 2012
B. The estimated beetle population for this species and area 7 years after 2012
C. The estimated percent decrease in the beetle population for this species and area each year after 2012
D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

37. The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

38. $$2|4 - x| + 3|4 - x| = 25$$

What is the positive solution to the given equation?

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39. $4a^2 + 20ab + 25b^2$

Which of the following is a factor of the polynomial above?

A. $a + b$
B. $2a + 5b$
C. $4a + 5b$
D. $4a + 25b$

40. The International Space Station orbits Earth at an average speed of 4.76 miles per second. What is the space station's average speed in miles per hour?

A. 285.6
B. 571.2
C. 856.8
D. 17136

41. Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is $\frac{3}{4}$ cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult's daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

A. $p = 0.5x$
B. $p = 5x$
C. $p = (0.05)^x$
D. $p = (1.05)^X$

42. A city has 50 city council members. A reporter polled a random sample of 20 city council members and found that 6 of those polled supported a specific bill. Based on the sample, which of the following is the best estimate of the number of city council members in the city who support the bill?

A. 6
B. 9
C. 15
D. 30

43. Residents of a town were surveyed to determine whether they are satisfied with the concession stand at the local park. A random sample of 200 residents was selected. All 200 responded, and 87% said they are satisfied. Based on this information, which of the following statements must be true?

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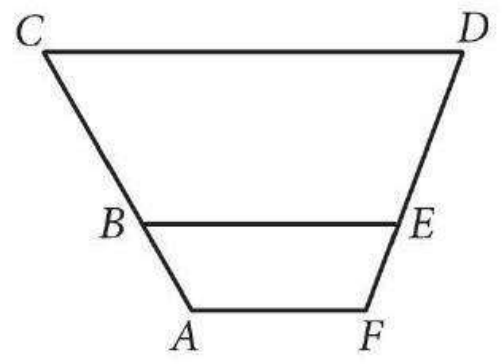
I. Of all the town residents, 87% would say they are satisfied with the concession stand at the local park.
II. If another random sample of 200 residents were surveyed, 87% would say they are satisfied.

A. Neither
B. I only
C. II only
D. I and II

44.



In the figure above, $\overline{AF}, \overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9, BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

A. 16.8
B. 17.5
C. 18.4
D. 19.6

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